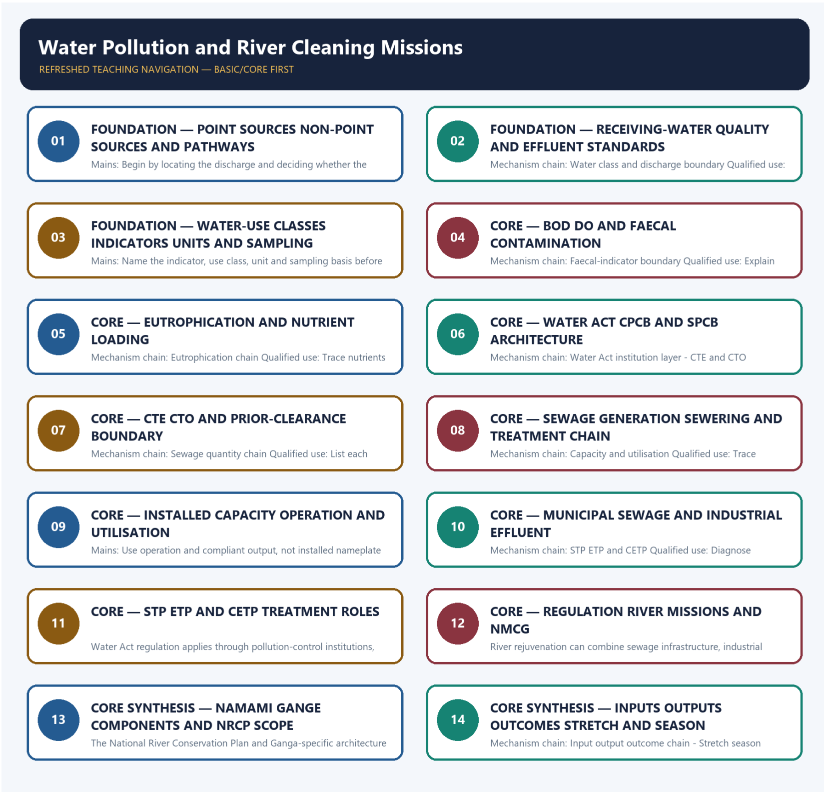
SOLVED PRACTICE WORKBOOK

Water Pollution and River Cleaning Missions - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Water Pollution and River Cleaning Missions - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Point and non-point sources?

A. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. B. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. C. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. D. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO.

Answer: A. Explanation: A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q2. Which option preserves the ecological boundary of Point and non-point sources?

A. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. B. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. C. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. D. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict.

Answer: B. Explanation: A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q3. Which statement uses Point and non-point sources without changing its scale, parameter or status?

A. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. B. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. C. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. D. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO.

Answer: C. Explanation: A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q4. Which option avoids the standard UPSC close-option trap about Point and non-point sources?

A. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. B. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. C. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. D. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different.

Answer: D. Explanation: A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q5. Which statement correctly identifies Water-quality and effluent boundary?

A. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. B. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. C. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. D. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO.

Answer: A. Explanation: Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q6. Which option preserves the ecological boundary of Water-quality and effluent boundary?

A. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. B. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. C. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. D. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict.

Answer: B. Explanation: Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q7. Which statement uses Water-quality and effluent boundary without changing its scale, parameter or status?

A. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. B. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. C. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. D. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict.

Answer: C. Explanation: Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status

categories.

Q8. Which option avoids the standard UPSC close-option trap about Water-quality and effluent boundary?

A. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. B. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. C. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. D. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other.

Answer: D. Explanation: Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q9. Which statement correctly identifies Water class and discharge boundary?

A. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. B. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. C. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. D. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution.

Answer: A. Explanation: A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q10. Which option preserves the ecological boundary of Water class and discharge boundary?

A. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. B. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. C. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. D. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution.

Answer: B. Explanation: A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q11. Which statement uses Water class and discharge boundary without changing its scale, parameter or status?

A. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. B. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. C. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. D. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges.

Answer: C. Explanation: A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q12. Which option avoids the standard UPSC close-option trap about Water class and discharge boundary?

A. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. B. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. C. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. D. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source.

Answer: D. Explanation: A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q13. Which statement correctly identifies BOD and DO relation?

A. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. B. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. C. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. D. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution.

Answer: A. Explanation: Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q14. Which option preserves the ecological boundary of BOD and DO relation?

A. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. B. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. C. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. D. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution.

Answer: B. Explanation: Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q15. Which statement uses BOD and DO relation without changing its scale, parameter or status?

A. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. B. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. C. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. D. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges.

Answer: C. Explanation: Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. The other options belong to different media, parameters, processes, scales, actors, institutions,

instruments or status categories.

Q16. Which option avoids the standard UPSC close-option trap about BOD and DO relation?

A. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. B. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. C. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. D. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO.

Answer: D. Explanation: Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q17. Which statement correctly identifies Faecal-indicator boundary?

A. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. B. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. C. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. D. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance.

Answer: A. Explanation: Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q18. Which option preserves the ecological boundary of Faecal-indicator boundary?

A. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. B. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. C. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. D. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance.

Answer: B. Explanation: Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q19. Which statement uses Faecal-indicator boundary without changing its scale, parameter or status?

A. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. B. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. C. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. D. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome.

Answer: C. Explanation: Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q20. Which option avoids the standard UPSC close-option trap about Faecal-indicator boundary?

A. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. B. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. C. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. D. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict.

Answer: D. Explanation: Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q21. Which statement correctly identifies Eutrophication chain?

A. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. B. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. C. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. D. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities.

Answer: A. Explanation: Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q22. Which option preserves the ecological boundary of Eutrophication chain?

A. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. B. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. C. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. D. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities.

Answer: B. Explanation: Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q23. Which statement uses Eutrophication chain without changing its scale, parameter or status?

A. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. B. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. C. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. D. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other.

Answer: C. Explanation: Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q24. Which option avoids the standard UPSC close-option trap about Eutrophication chain?

A. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. B. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. C. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. D. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution.

Answer: D. Explanation: Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q25. Which statement correctly identifies Water Act institution layer?

A. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. B. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. C. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. D. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance.

Answer: A. Explanation: The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q26. Which option preserves the ecological boundary of Water Act institution layer?

A. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. B. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. C. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. D. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other.

Answer: B. Explanation: The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q27. Which statement uses Water Act institution layer without changing its scale, parameter or status?

A. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. B. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. C. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. D. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other.

Answer: C. Explanation: The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q28. Which option avoids the standard UPSC close-option trap about Water Act institution layer?

A. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. B. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. C. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. D. The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges.

Answer: D. Explanation: The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q29. Which statement correctly identifies CTE and CTO boundary?

A. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. B. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. C. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. D. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities.

Answer: A. Explanation: Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q30. Which option preserves the ecological boundary of CTE and CTO boundary?

A. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. B. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. C. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. D. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other.

Answer: B. Explanation: Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q31. Which statement uses CTE and CTO boundary without changing its scale, parameter or status?

A. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. B. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. C. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. D. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation.

Answer: C. Explanation: Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q32. Which option avoids the standard UPSC close-option trap about CTE and CTO boundary?

A. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. B. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. C. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. D. Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance.

Answer: D. Explanation: Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q33. Which statement correctly identifies Sewage quantity chain?

A. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. B. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. C. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. D. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other.

Answer: A. Explanation: Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q34. Which option preserves the ecological boundary of Sewage quantity chain?

A. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. B. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. C. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. D. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation.

Answer: B. Explanation: Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q35. Which statement uses Sewage quantity chain without changing its scale, parameter or status?

A. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. B. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. C. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. D. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement.

Answer: C. Explanation: Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q36. Which option avoids the standard UPSC close-option trap about Sewage quantity chain?

A. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. B. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. C. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. D. Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities.

Answer: D. Explanation: Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q37. Which statement correctly identifies Capacity and utilisation?

A. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. B. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. C. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. D. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation.

Answer: A. Explanation: An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q38. Which option preserves the ecological boundary of Capacity and utilisation?

A. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. B. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. C. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. D. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation.

Answer: B. Explanation: An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q39. Which statement uses Capacity and utilisation without changing its scale, parameter or status?

A. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. B. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. C. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. D. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement.

Answer: C. Explanation: An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q40. Which option avoids the standard UPSC close-option trap about Capacity and utilisation?

A. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. B. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. C. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. D. An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome.

Answer: D. Explanation: An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q41. Which statement correctly identifies Municipal and industrial streams?

A. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. B. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. C. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. D. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement.

Answer: A. Explanation: Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q42. Which option preserves the ecological boundary of Municipal and industrial streams?

A. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. B. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. C. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. D. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement.

Answer: B. Explanation: Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q43. Which statement uses Municipal and industrial streams without changing its scale, parameter or status?

A. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. B. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. C. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. D. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river.

Answer: C. Explanation: Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q44. Which option avoids the standard UPSC close-option trap about Municipal and industrial streams?

A. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. B. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. C. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. D. Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other.

Answer: D. Explanation: Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q45. Which statement correctly identifies STP ETP and CETP?

A. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. B. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. C. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. D. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement.

Answer: A. Explanation: An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q46. Which option preserves the ecological boundary of STP ETP and CETP?

A. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. B. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. C. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. D. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river.

Answer: B. Explanation: An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q47. Which statement uses STP ETP and CETP without changing its scale, parameter or status?

A. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. B. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. C. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. D. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages.

Answer: C. Explanation: An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q48. Which option avoids the standard UPSC close-option trap about STP ETP and CETP?

A. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. B. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. C. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. D. An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation.

Answer: D. Explanation: An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q49. Which statement correctly identifies Regulation and mission boundary?

A. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. B. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. C. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. D. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement.

Answer: A. Explanation: Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q50. Which option preserves the ecological boundary of Regulation and mission boundary?

A. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. B. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. C. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. D. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river.

Answer: B. Explanation: Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q51. Which statement uses Regulation and mission boundary without changing its scale, parameter or status?

A. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. B. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. C. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. D. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river.

Answer: C. Explanation: Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q52. Which option avoids the standard UPSC close-option trap about Regulation and mission boundary?

A. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. B. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. C. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. D. Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement.

Answer: D. Explanation: Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q53. Which statement correctly identifies NMCG institutional boundary?

A. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. B. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. C. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. D. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement.

Answer: A. Explanation: NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q54. Which option preserves the ecological boundary of NMCG institutional boundary?

A. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. B. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. C. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. D. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages.

Answer: B. Explanation: NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q55. Which statement uses NMCG institutional boundary without changing its scale, parameter or status?

A. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. B. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. C. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. D. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean.

Answer: C. Explanation: NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q56. Which option avoids the standard UPSC close-option trap about NMCG institutional boundary?

A. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. B. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. C. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. D. NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator.

Answer: D. Explanation: NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q57. Which statement correctly identifies Namami Gange component boundary?

A. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. B. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. C. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. D. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean.

Answer: A. Explanation: River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q58. Which option preserves the ecological boundary of Namami Gange component boundary?

A. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. B. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. C. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. D. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean.

Answer: B. Explanation: River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q59. Which statement uses Namami Gange component boundary without changing its scale, parameter or status?

A. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. B. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. C. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. D. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean.

Answer: C. Explanation: River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. The other options belong to different media, parameters, processes, scales,

actors, institutions, instruments or status categories.

Q60. Which option avoids the standard UPSC close-option trap about Namami Gange component boundary?

A. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. B. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. C. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. D. River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement.

Answer: D. Explanation: River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q61. Which statement correctly identifies NRCP and river scope?

A. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. B. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. C. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. D. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages.

Answer: A. Explanation: The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q62. Which option preserves the ecological boundary of NRCP and river scope?

A. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. B. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. C. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. D. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean.

Answer: B. Explanation: The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q63. Which statement uses NRCP and river scope without changing its scale, parameter or status?

A. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. B. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. C. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. D. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different.

Answer: C. Explanation: The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. The other

options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.**

Q64. Which option avoids the standard UPSC close-option trap about NRCP and river scope?

A. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. B. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. C. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. D. The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river.

Answer: D. Explanation: The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q65. Which statement correctly identifies Input output outcome chain?

A. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. B. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. C. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. D. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution.

Answer: A. Explanation: Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q66. Which option preserves the ecological boundary of Input output outcome chain?

A. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. B. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. C. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. D. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different.

Answer: B. Explanation: Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q67. Which statement uses Input output outcome chain without changing its scale, parameter or status?

A. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. B. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. C. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. D. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values.

Answer: C. Explanation: Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q68. Which option avoids the standard UPSC close-option trap about Input output outcome chain?

A. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. B. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. C. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. D. Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages.

Answer: D. Explanation: Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q69. Which statement correctly identifies Stretch season indicator?

A. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. B. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. C. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. D. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different.

Answer: A. Explanation: River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q70. Which option preserves the ecological boundary of Stretch season indicator?

A. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. B. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. C. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. D. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other.

Answer: B. Explanation: River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q71. Which statement uses Stretch season indicator without changing its scale, parameter or status?

A. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. B. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. C. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. D. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different.

Answer: C. Explanation: River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q72. Which option avoids the standard UPSC close-option trap about Stretch season indicator?

A. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. B. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. C. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. D. River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean.

Answer: D. Explanation: River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q73. Which statement correctly identifies Dilution and ecological flow?

A. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. B. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. C. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. D. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other.

Answer: A. Explanation: Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q74. Which option preserves the ecological boundary of Dilution and ecological flow?

A. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. B. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. C. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. D. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source.

Answer: B. Explanation: Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q75. Which statement uses Dilution and ecological flow without changing its scale, parameter or status?

A. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. B. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. C. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. D. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other.

Answer: C. Explanation: Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q76. Which option avoids the standard UPSC close-option trap about Dilution and ecological flow?

A. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. B. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. C. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. D. Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution.

Answer: D. Explanation: Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q77. Which statement correctly identifies Audited evidence boundary?

A. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. B. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. C. A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. D. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other.

Answer: A. Explanation: Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q78. Which option preserves the ecological boundary of Audited evidence boundary?

A. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. B. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. C. Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. D. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO.

Answer: B. Explanation: Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q79. Which statement uses Audited evidence boundary without changing its scale, parameter or status?

A. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. B. A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. C. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. D. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO.

Answer: C. Explanation: Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. The other options belong to different media, parameters,

processes, scales, actors, institutions, instruments or status categories.

Q80. Which option avoids the standard UPSC close-option trap about Audited evidence boundary?

A. Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. B. Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. C. Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. D. Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values.

Answer: D. Explanation: Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

PYQS AND ANSWER PRACTICE

AUDITED WATER-POLLUTION, TREATMENT AND RIVER-MISSION PYQ OWNERSHIP

Audited ledgers route direct Mains demands on industrial river pollution and freshwater technologies, plus objective concepts on membrane bioreactors, activated carbon, PFAS, microbeads and sand-mining effects. No objective key or current mission metric is inferred.

OWNER PYQ LEDGER EXTRACTS

9. PYQ application

- FACT: **2024 GS-III direct PYQ:** “Industrial pollution of river water is a significant environmental issue in India.” It asks for mitigation measures and government initiatives. Route the exact demand via `../README.md`.
- FACT: **2024 GS-III direct PYQ (250 words):** “The world is facing an acute shortage of clean and safe freshwater. What are the alternative technologies which can solve this crisis? Briefly discuss any three such technologies citing their key merits and demerits.” ANALYSIS: Note the exact demand: **three** technologies, each with **merits and demerits** - a structured three-block answer (e.g., desalination, wastewater recycling/reuse, atmospheric water generation or advanced membrane treatment), not a general water-scarcity essay.
- FACT: **2025 GS-III direct PYQ (250 words):** “Examine the factors responsible for depleting groundwater in India. What are the steps taken by the government to mitigate such depletion of groundwater?” ANALYSIS: This is a **groundwater quantity** question; keep it distinct from the surface-water **quality** framing of Namami Gange, and use Atal Bhujal Yojana, Jal Shakti Abhiyan/Catch the Rain, the Central Ground Water Authority's notified over-exploited blocks and the Master Plan for Artificial Recharge as the "steps taken".
- ANALYSIS: Recurring Prelims pattern: identify the correct water-quality indicator (BOD/DO/ faecal coliform) implied by a described pollution scenario.
- ANALYSIS: Mains linkage: the sewage-versus-industrial-effluent pollution-source distinction is used to argue for infrastructure investment (sewage treatment) as the primary river- cleaning priority.

Recent PYQ Integration (2024-2025)

> **Status:** 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2024-2025.md`, `_PYQ-ROUTING-PRELIMS-2024-2025.md`. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** GS-III, Prelims GS-I
- **Routed question demands:** 5

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	GS-III	7	Industrial pollution of river water and mitigation measures	Discuss · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2024	GS-III	15	Alternative technologies for the freshwater crisis	Discuss · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2024	Prelims GS-I	17	PFAS in consumer products	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2024	Prelims GS-I	39	'Membrane Bioreactors' in wastewater treatment	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	Prelims GS-I	50	Activated carbon for removing pollutants from effluents	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Industrial pollution of river water and mitigation measures
- Alternative technologies for the freshwater crisis
- PFAS in consumer products
- 'Membrane Bioreactors' in wastewater treatment
- Activated carbon for removing pollutants from effluents

> This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-PRELIMS-2018-2023.md`. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	Prelims GS-I	81	Heavy sand mining riverbeds environmental and groundwater consequences	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	Prelims GS-I	30	Environmental concern about microbeads released in water	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Heavy sand mining riverbeds environmental and groundwater consequences
- Environmental concern about microbeads released in water

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

10. PYQ-based analytical application

- FACT: **2024 GS-III direct PYQ:** industrial river-water pollution-mitigation

measures plus government initiatives. Use source segregation, consent/ monitoring, treatment, liability and basin governance; exact route: `../README.md`.

- ANALYSIS: Prelims questions on Namami Gange's institutional home (Ministry of Jal Shakli/NMCG) and

its multi-component design should be answered by recalling the full component list, not just the infrastructure element.

- ANALYSIS: Mains answers on "river cleaning in India" should explicitly engage the "last-mile" gap

and the stretch/season-specific water-quality-claim caution to demonstrate analytical rigour beyond a simple programme description.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish receiving-water criteria from source-effluent standards. Answer in about 150 words.

Model thesis: Claim: Point and non-point sources. **Named evidence/example:** A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water-quality and effluent boundary. **Named evidence/example:** Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water class and discharge boundary. **Named evidence/example:** A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different.
- Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other.
- A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source.

Qualified conclusion: Claim: Point and non-point sources. **Named evidence/example:** A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water-quality and effluent boundary. **Named evidence/example:** Receiving-water quality describes the water body, whereas an effluent standard controls a discharge from a source; meeting one cannot be assumed from the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water class and discharge boundary. **Named evidence/example:** A designated water-use class or criterion is not an industry effluent limit; each numeric value must retain its indicator, unit, sampling basis and legal source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain the use and limits of BOD, DO and faecal indicators. Answer in about 150 words.

Model thesis: Claim: BOD and DO relation. **Named evidence/example:** Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Faecal-indicator boundary. **Named evidence/example:** Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Eutrophication chain. **Named evidence/example:** Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO.
- Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict.
- Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution.

Qualified conclusion: Claim: BOD and DO relation. **Named evidence/example:** Biochemical Oxygen Demand indicates oxygen used in microbial decomposition of organic matter, while Dissolved Oxygen is oxygen available in water; high organic load can raise BOD and depress DO. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Faecal-indicator boundary. **Named evidence/example:** Faecal coliform is an indicator of faecal contamination and possible pathogen risk; it is not a direct count of every pathogen or a complete water-quality verdict. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Eutrophication chain. **Named evidence/example:** Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 3 - 15 MARKS

Question: Trace sewage from generation to compliant treatment and reuse. Answer in about 250 words.

Model thesis: Claim: Sewage quantity chain. **Named evidence/example:** Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity and utilisation. **Named evidence/example:** An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** STP ETP and CETP. **Named evidence/example:** An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities.
- An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome.
- An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation.

Qualified conclusion: Claim: Sewage quantity chain. **Named evidence/example:** Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity and utilisation. **Named evidence/example:** An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** STP ETP and CETP. **Named evidence/example:** An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 4 - 15 MARKS

Question: Distinguish Water Act regulation from river-mission implementation. Answer in about 250 words.

Model thesis: Claim: Water Act institution layer. **Named evidence/example:** The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** CTE and CTO boundary. **Named evidence/example:** Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Regulation and mission boundary. **Named evidence/example:** Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NMCG institutional boundary. **Named evidence/example:** NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NRCP and river scope. **Named evidence/example:** The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges.
- Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance.
- Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement.
- NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator.
- The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river.

Qualified conclusion: Claim: Water Act institution layer. **Named evidence/example:** The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** CTE and CTO boundary. **Named evidence/example:** Consent to Establish and Consent to Operate regulate a source under pollution-control law; neither is the same as prior environmental clearance, river-mission approval or proof of continuous compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Regulation and mission boundary. **Named evidence/example:** Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NMCG institutional boundary. **Named evidence/example:** NMCG implements the Ganga mission within its notified institutional architecture; it is not interchangeable with CPCB, an SPCB, an urban local body or a generic all-river regulator. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NRCP and river scope. **Named evidence/example:** The National River Conservation Plan and Ganga-specific architecture have different programme scopes; a Ganga institution or result cannot automatically be assigned to every river. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate river cleaning through input, output, utilisation and outcome metrics. Answer in about 300 words.

Model thesis: Claim: Sewage quantity chain. **Named evidence/example:** Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity and utilisation. **Named evidence/example:** An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Namami Gange component boundary. **Named evidence/example:** River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Input output outcome chain. **Named evidence/example:** Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Stretch season indicator. **Named evidence/example:** River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Dilution and ecological flow. **Named evidence/example:** Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities.
- An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome.
- River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement.
- Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages.
- River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean.
- Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution.

Qualified conclusion: Claim: Sewage quantity chain. **Named evidence/example:** Sewage generated, sewered flow, installed treatment capacity, commissioned capacity, actual inflow, compliant treatment and reuse are separate quantities. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity and utilisation. **Named evidence/example:** An STP's rated or installed capacity is not its actual operating load, treatment performance, utilisation or receiving-river outcome. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Namami Gange component boundary. **Named evidence/example:** River rejuvenation can combine sewage infrastructure, industrial monitoring, river-surface action, biodiversity, afforestation and public participation; visible works alone do not prove water-quality improvement. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Input output outcome chain. **Named evidence/example:** Mission outlay, sanction, expenditure, infrastructure completed, flow treated, pollutant load reduced and river-quality outcome are distinct stages. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Stretch season indicator. **Named evidence/example:** River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Dilution and ecological flow. **Named evidence/example:** Higher flow may dilute a measured concentration without removing pollutant mass or source discharge; river rejuvenation therefore cannot be reduced to dilution. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 6 - 20 MARKS

Question: Build an integrated response to municipal, industrial and diffuse water pollution. Answer in about 300 words.

Model thesis: Claim: Point and non-point sources. **Named evidence/example:** A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Eutrophication chain. **Named evidence/example:** Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water Act institution layer. **Named evidence/example:** The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Municipal and industrial streams. **Named evidence/example:** Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** STP ETP and CETP. **Named evidence/example:** An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Regulation and mission boundary. **Named evidence/example:** Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Stretch season indicator. **Named evidence/example:** River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Audited evidence boundary. **Named evidence/example:** Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different.
- Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution.

- The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges.
- Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other.
- An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation.
- Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement.
- River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean.
- Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values.

Qualified conclusion: Claim: Point and non-point sources. **Named evidence/example:** A point source has an identifiable discharge location, while non-point pollution is diffuse across a catchment; the monitoring and regulatory tools are therefore different. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Eutrophication chain. **Named evidence/example:** Excess nutrient loading can drive algal growth, decomposition and oxygen depletion; nutrient enrichment is distinct from, though it can coexist with, sewage and toxic industrial pollution. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water Act institution layer. **Named evidence/example:** The Water Act establishes the CPCB-SPCB pollution-control architecture, while state boards administer major consent, monitoring and enforcement functions for discharges. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Municipal and industrial streams. **Named evidence/example:** Municipal sewage and industrial effluent differ in source, composition, treatment train, monitoring and responsible institution; neither should be used as a proxy for the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** STP ETP and CETP. **Named evidence/example:** An STP treats sewage, an ETP treats an individual establishment's effluent, and a CETP serves a group of units; installation is not proof of compliant operation. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Regulation and mission boundary. **Named evidence/example:** Water Act regulation applies through pollution-control institutions, while a river-cleaning mission coordinates projects and basin action; a mission does not replace statutory consent enforcement. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Stretch season indicator. **Named evidence/example:** River quality varies by monitoring location, season, flow and indicator; a result for one stretch or period cannot establish that an entire river is clean. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Audited evidence boundary. **Named evidence/example:** Audited ledgers carry industrial river pollution, freshwater treatment technologies, membrane bioreactors, activated carbon, microbeads and sand-mining effects into practice without inventing keys, capacities, outlays or river-quality values. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.